A survey on Metadata for Semantic and Data Mappings

You are invited to participate in this study on Metadata for semantic and data mappings. This study aims to identify a set of metadata fields that can be used to annotate semantic and data mappings to facilitate their discovery, reusability, and quality assessment. This study is being conducted as part of the requirements for the Ph.D. in computer science at Trinity College Dublin.

Procedures of this study:
In this survey, you will be asked to describe your current level of data and semantic mapping experience; then, you will be asked to rate the relevance of various metadata fields to two mapping-related tasks.
There will be separate sections for each of these tasks. After reading and rating predefined metadata sets relevant to these tasks, a question will ask whether you suggest any additional metadata fields beyond those suggested by the researcher. It is expected that your participation in this survey will not exceed 15 minutes.

Publication:
We plan to publish the results of our research in academic journals and conference proceedings. Also, the research results will be published in a Ph.D. dissertation at Trinity College Dublin. All participants in this study will be anonymous, and the findings will be published without identifying you or anyone else.

Please review the following information sheet (shorturl.at/lxAOY) **for further details regarding the study which will assist you in making an informed decision on whether to participate**.

Lead Researcher:
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Declaration:
• I am 18 years or older and am competent to provide consent.
• I have read a document providing information about this research and this consent form. I have had the opportunity to ask questions, and all my questions have been answered to my satisfaction, and I understand the description of the research that is being provided to me.
• I agree that my data is used for scientific purposes and I have no objection that my data is published in scientific publications in a way that does not reveal my identity.
• I understand that if I make illicit activities known, these will be reported to appropriate authorities.

• I understand that I may refuse to answer any question and that I may withdraw at any time without penalty.
• I understand that if my data has been fully anonymized so that it can no longer be attributed to me, then it will no longer be possible to withdraw
• I freely and voluntarily agree to be part of this research study, though without prejudice to my legal and ethical rights.
• I understand that my participation is fully anonymous and that no personal details about me will be recorded.
• I understand that personal information about me, including the transfer of this personal information about me outside of the EU, will be protected in accordance with the General Data Protection Regulation.
• I have received a copy of this agreement.

* Indicates required question

1. Please indicate your consent before proceeding *

Mark only one oval.

- ( ) I consent to participate in this study, and consent to the data processing necessary to enable my participation and to achieve the research goals of this study. *Skip to question 2*
- ( ) I don't consent

A survey on Metadata for Semantic and Data Mappings

1- **Introduction**

In the linked data and semantic web world, mappings are crucial to overcoming semantics heterogeneity, which is one of the major barriers to interoperability. Mapping can be defined as the process of specifying how two semantic or data models relate to each other. It can be categorized into three main types: ontologies mapping, interlinking, and uplift/downlift mapping.

- The uplift mapping process involves converting different data formats to RDF
- The downlift mapping process refers to the reverse of the uplift mapping process.
- Interlinking is a mapping applied within or between datasets of RDF in order to connect them.
- Ontologies mapping involves mapping the vocabularies of two ontologies.

Developing these mappings is not an easy process, since they are intended for a specific purpose and require expert knowledge, resources, and time. To facilitate its development and reduce the associated cost, the re-use of previously developed mappings is considered necessary. It is put forward that annotating mappings with metadata may help in improving mapping discovery, reuse, and quality assessment. As a part of this research, we are attempting to create an approach that will enable the annotation of mappings with metadata across the complete mapping development lifecycle.

2- **An overview of the survey structure**:

There are two sections in the survey in which you can indicate how relevant a predefined set of metadata fields are to two different mapping-related tasks:

- Find mapping that can be reused
- Evaluate the quality of mapping

These metadata fields were identified based on a proposed mapping lifecycle. The mapping lifecycle describes the stages involved in the construction of these mappings, beginning with the analysis of their purpose, and continuing to their management over time.

As one of the objectives of this study is to determine whether prior experience influences a person's assessment of the relevancy of metadata fields to mapping-related tasks, it is necessary for you to answer the following question regarding your experience before beginning this survey:

2. How would you describe your current level of experience in regards to data and semantic mapping *

Mark only one oval.

- No experience
- Beginner - "Beginner refers to someone who has been introduced to data and semantic mappings but has not applied them to a project or has less than 6 months' experience"
- Intermediate - "Intermediate refers to someone who has at least six months experience working on real world projects and reasonable experience with data and semantic mappings"
- Expert - "Expert refers to someone who have many years of experience working on real-world projects and researching data and semantic mappings"

1- Find a mapping that can be re-used

The process of creating mappings involves a considerable amount of time, effort, and expertise. This can be mitigated by reusing previously developed mappings. However, looking for mappings before reusing them is an essential step. In the event that mappings are stored in one portal, reasoning over the metadata fields associated with the mappings is one means of finding appropriate mappings to reuse.

In this section, you will be asked based on your experience to rate the relevance of metadata fields to the mapping discovery process for the purpose of reuse. It is expected that relevant metadata fields will facilitate the discovery process when queried.

On scale of 1 to 5, Please rate the following metadata fields (in your opinion) based on how relevant they are to the mapping discovery process.

3. Metadata about stakeholders (A stakeholder is anybody who has contributed to the development of a mapping, such as knowledge engineers and domain experts)

Mark only one oval per row.

	1 Highly relevant to mapping discovery	2	3	4	5 Irrelevant to mapping discovery
Stakeholder's Human Readable Name	⬭	⬭	⬭	⬭	⬭
Stakeholder's URI	⬭	⬭	⬭	⬭	⬭
Role of stakeholder at each stage of the mapping process	⬭	⬭	⬭	⬭	⬭
Background on the stakeholder	⬭	⬭	⬭	⬭	⬭
Stakeholder organization	⬭	⬭	⬭	⬭	⬭

4. Metadata related to the purpose of the mapping project

Mark only one oval per row.

	1 Highly relevant to mapping discovery	2	3	4	5 Irrelevant to mapping discovery
Mapping requirements	◯	◯	◯	◯	◯
Type of mapping (ontologies mapping, uplift mapping, interlinking)	◯	◯	◯	◯	◯
Mapping domain (Medical, Library ...etc)	◯	◯	◯	◯	◯
Domain assumptions	◯	◯	◯	◯	◯
Any technical requirements	◯	◯	◯	◯	◯
Risks or issues identified	◯	◯	◯	◯	◯

identified prior to development

prior to development

5. Metadata to describe inputs that will be mapped (In the case of ontology mapping, inputs would be ontologies, datasets or entities for the interlinking type of mapping, and any source of data for uplift mapping.)

Mark only one oval per row.

	1 Highly relevant to mapping discovery	2	3	4	5 Irrelevant to mapping discovery
Input name in human readable form	◯	◯	◯	◯	◯
Input URI (Uniform Resource Identifier)	◯	◯	◯	◯	◯
Source of this input	◯	◯	◯	◯	◯
Input type (eg: ontology , RDF dataset ..etc)	◯	◯	◯	◯	◯
Input creator	◯	◯	◯	◯	◯

Input format (eg: csv, xml ..etc)					

6. Metadata to describe the design process of the mapping. . During this process, a decision is made as to whether an existing mapping can be reused or if a new one should be developed.

Mark only one oval per row.

	1 Highly relevant to mapping discovery	2	3	4	5 Irrelevant to mapping discovery
Final design decisions (create new mapping, reuse existing mapping, integrate multiple mapping ..etc)	◯	◯	◯	◯	◯
Design decision justification	◯	◯	◯	◯	◯
Mapping quality metrics (metrics to consider during the development)	◯	◯	◯	◯	◯

7. Metadata to describe the development process of a mapping

Mark only one oval per row.

	1 Highly relevant to mapping discovery	2	3	4	5 Irrelevant to mapping discovery
Mapping Human readable name	⬭	⬭	⬭	⬭	⬭
Mapping URI (Uniform Resource Identifier)	⬭	⬭	⬭	⬭	⬭
Development of mapping started on (date/time)	⬭	⬭	⬭	⬭	⬭
Development of mapping end on (date/time)	⬭	⬭	⬭	⬭	⬭
Tools used during the development	⬭	⬭	⬭	⬭	⬭
Mapping method					

(manual - automatic) Mapping method					
(manual - automatic) Mapping algorithm					
Mapping algorithm Mapping format					
Mapping format					

8. Metadata for describing the testing process of a mapping

Mark only one oval per row.

	1 Highly relevant to mapping discovery	2	3	4	5 Irrelevant to mapping discovery
Testing unique identifier	⬭	⬭	⬭	⬭	⬭
Testing name	⬭	⬭	⬭	⬭	⬭
Testing Type (Validation, Verification, Quality testing)	⬭	⬭	⬭	⬭	⬭
Testing Date/Time	⬭	⬭	⬭	⬭	⬭
Testing Result	⬭	⬭	⬭	⬭	⬭

9. Metadata for describing the management and maintenance of mappings

Mark only one oval per row.

	1 Highly relevant to mapping discovery	2	3	4	5 Irrelevant to mapping discovery
Publisher name					
Publisher source					
Version number					
Version data/time					

10. Please let us know of any other metadata fields you recommend adding to improve the mapping discovery process

2- Evaluating the quality of mapping

Evaluating the quality of mapping is one of mapping related tasks. During this process, mapping quality should be assessed based on predefined quality metrics or criteria.

In this section, you will be asked based on your experience to rate the relevance of metadata fields to the quality assessment process. It is expected that relevant metadata fields will facilitate the quality assessment process when queried.

On scale of 1 to 5, Please rate the following metadata fields (in your opinion) based on how relevant they are to the mapping quality assessment process.

11. Metadata about stakeholders (A stakeholder is anybody who has contributed to the development of a mapping, such as knowledge engineers and domain experts)

Mark only one oval per row.

	1 Highly relevant to quality assessment	2	3	4	5 Irrelevant to quality assessment
Stakeholder's Human Readable Name	◯	◯	◯	◯	◯
Stakeholder's URI	◯	◯	◯	◯	◯
Role of stakeholder at each stage of the mapping process	◯	◯	◯	◯	◯
Background on the stakeholder	◯	◯	◯	◯	◯
Stakeholder organization	◯	◯	◯	◯	◯

12. Metadata related to the purpose of this mapping project

Mark only one oval per row.

	1 Highly relevant to quality assessment	2	3	4	5 Irrelevant to quality assessment
Mapping requirements	⬭	⬭	⬭	⬭	⬭
Type of mapping (ontologies mapping, uplift mapping, interlinking)	⬭	⬭	⬭	⬭	⬭
Mapping domain (Medical, Library...etc)	⬭	⬭	⬭	⬭	⬭
Domain assumptions	⬭	⬭	⬭	⬭	⬭
Any technical requirements	⬭	⬭	⬭	⬭	⬭
Risks or issues identified prior to	⬭	⬭	⬭	⬭	⬭

prior to development

development

13. Metadata to describe inputs that will be mapped (In the case of ontology mapping, inputs would be ontologies, datasets or entities for the interlinking type of mapping, and any source of data for uplift mapping.)

Mark only one oval per row.

	1 Highly relevant to quality assessment	2	3	4	5 Irrelevant to quality assessment
Input name in human readable form	◯	◯	◯	◯	◯
Input URI (Uniform Resource Identifier)	◯	◯	◯	◯	◯
Source of this input	◯	◯	◯	◯	◯
Input type (eg: ontology , RDF dataset ..etc)	◯	◯	◯	◯	◯
Input creator	◯	◯	◯	◯	◯
Input					

Input format (eg: csv, xml ..etc)					

14. Metadata to describe the design process of the mapping. . During this process, a decision is made as to whether an existing mapping can be reused or if a new one should be developed.

Mark only one oval per row.

	1 Highly relevant to quality assessment	2	3	4	5 Irrelevant to quality assessment
Final design decisions (create new mapping, reuse existing mapping, integrate multiple mapping ..etc)	◯	◯	◯	◯	◯
Design decision justification	◯	◯	◯	◯	◯
Mapping quality metrics (metrics to consider during the development)	◯	◯	◯	◯	◯

15. Metadata to describe the development process of a mapping

Mark only one oval per row.

	1 Highly relevant to quality assessment	2	3	4	5 Irrelevant to quality assessment
Mapping Human readable name					
Mapping URI (Uniform Resource Identifier)					
Development of mapping started on (date/time)					
Development of mapping end on (date/time)					
Tools used during the development					
Mapping method (manual -					

(manual - automatic)					
Mapping algorithm					
Mapping format					

16. Metadata for describing the testing process of a mapping

Mark only one oval per row.

	1 Highly relevant to quality assessment	2	3	4	5 Irrelevant to quality assessment
Testing Unique identifier	◯	◯	◯	◯	◯
Testing name	◯	◯	◯	◯	◯
Testing Type (Validation, Verification, Quality testing)	◯	◯	◯	◯	◯
Testing Date/Time	◯	◯	◯	◯	◯
Testing Result	◯	◯	◯	◯	◯

17. Metadata for describing the management and maintenance of mappings

Mark only one oval per row.

	1 Highly relevant to quality assessment	2	3	4	5 Irrelevant to quality assessment
Publisher name					
Publisher source					
Version number					
Version data/time					

18. Finally, if you are aware of any other metadata fields that would improve assessing the quality of mappings, please let us know about them.